A revamp, of sorts

REMEMBER THE VERY BASIC DIAGRAM OF HOW A COMPUTER WORKS? We were taught that in the very first 'introduction to computers' classes back in school. For me it's been more than 25 years but the diagram is still fresh in my mind – three rectangles placed side-by-side from left to right with little interconnects between them. The first being the input, the last being the output and the middle rectangle would be further sectioned into CPU and Memory. The CPU would again be sectioned into Control and the ALU. Minor variations of this diagram have been taught to practically everyone who's gone through a formal education system but not a lot of people know it as the Von Neumann architecture. This was introduced to the world in 1945. That was 77 years ago. And now, we might be looking at completely changing the way we organise computers.

It's remarkable that the Von Neumann architecture has remained valid for so many decades. Much like how Moore's Law has more or less been applicable for transistor density after all these years. And now, just like Moore's Law, even the Von Neumann architecture is being relooked at for an overhaul.

The problem is AI. And to be more specific, the amount of data that AI needs to be trained properly. The Von Neumann architecture was designed with the assumption that all the elements of computing would scale at more or less the rate over time. The processing capability of the computing cores within processors have grown by leaps and bounds and the buses or the interfaces that do the job of moving data to and fro from the storage have also scaled tremendously. We've seen different types of computing cores being designed as well. Some for very specific purposes and others being very general-purpose. The desktops and laptops that we use have these general-purpose cores whereas the graphics cards and FPGAs are very purpose-built. The Von Neumann architecture is how all of these devices were designed. Even ASICs, which are an extreme version of purpose-built computers, still follow the Von Neumann architecture. Every workload that needed a computer to handle, did not require folks to rethink this very basic design. And then came AI...

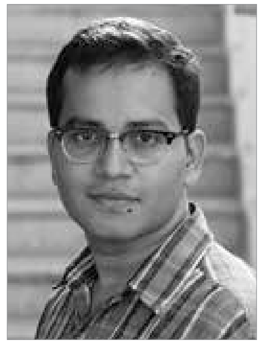
AI workloads are heavily parallelised tasks that need to be executed rapidly. Whether you're training an AI model or deploying it into production, the amount of data that it needs to churn so that it can generate an output is tremendous. Especially when the data being fed are images and videos which are inherently massive in size compared to plain text. Either the buses that move the data around need to speed up or data needs to move closer to the computing cores. Both approaches can be realised using the Von Neumann architecture.

Purpose-built AI training servers come with ridiculous amounts of memory and super-fast interfaces. For example, the current gen NVIDIA DGX H100 system, an extremely popular system for training and deploying AI, comes with 2 TB of system memory and the bandwidth of the interfaces is up to 400 GB/s. The average PC comes with 8 GB of memory and 32 or 64 GB/s interfaces. It's not that the speed or the capacity isn't enough, it's the cost of fetching data from the memory to the computing cores.

Moving two pieces of data costs close to 700 times the energy as multiplying the very same two pieces. With small amounts of data, this 700x factor is barely noticeable. But when you're training AI models, then a ridiculous amount of energy is spent just in fetching data and sending it back. This is where the compute cores, that everyone likes to harp about, needs to take a back seat. And the memory takes centrestage. In Memory Computing, the next major step in computing will have compute cores and memory getting clubbed together.

Some of these approaches are already available. Samsung has merged their compute cores with memory and they're calling it PIM. And then there are companies such as Mythic which have processors with massive amounts of memory within the SoC. Both of these approaches leave very little scope for other compute cores to be attached. We could see a form of vendor lock-ins creeping into the computers being built for AI. Although, it will be ages before the normal laptops and desktops that we use get affected in any way.

Sure, the 'AI' processing element within processors might change a little or we might see memory chips being tacked onto the processors but the Von Neumann architecture will very much be valid. d



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